Project Proposal

Team Members

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Proposal #1: StudyByte

Proposal #2: UniMart

Proposal #1: StudyByte

A Peer-to-Peer Micro-Tutoring Marketplace

This project introduces a micro-tutoring Information System that facilitates student-to-student knowledge exchange. Unlike traditional tutoring services, our platform focuses on **on-demand topic mastery** rather than full-course enrollment. Integrated with a custom digital wallet and escrow system, it allows student tutors to earn 'Study Tokens' that are redeemable for real-world currency upon reaching performance milestones

Project Objectives

- Enable student-to-student micro-tutoring
- Support topic-based, on-demand learning
- Reward tutors using a digital token system
- Ensure secure transactions with escrow
- Promote quality through performance milestones

Literature Review

1. Online learning platforms (Coursera, Udemy).
2. Freelancing and Tutoring websites.

User Roles

- **Student (Learner):** Searches for specific topics, buys sessions using tokens, and rates tutors.
- **Student (Tutor):** Lists expertise by topic, sets "token price" per session, and manages their availability.
- **Admin (University/Staff):** Oversees the marketplace, handles disputes, and manages the "Token-to-Cash" conversion requests.

Core Features

A. The “MarketPlace” Module

- **Topic Catalog:** Instead of "Calculus 101," think "Integration by Parts" or "Python Loops."
- **Search & Filter:** Filter by department, price (tokens), rating, or "Urgency" (e.g., help needed before a midterm).
- **Booking System:** A calendar integration where learners can see a tutor's free slots.

B. The Token Economy

- **Wallet Management:** Real-time balance updates for users.
- **Escrow Service:** When a session is booked, tokens are held by the system and only released to the tutor once the learner confirms the session happened.
- **Conversion Portal:** A dashboard for tutors to see how close they are to the "Cash-out" threshold.

C. Interaction and Validation

- **Session Proof:** A generated "Meeting Code" that the learner gives the tutor at the end of the session to trigger the payment.
- **Rating & Reviews:** Social proof to ensure quality control on campus.

Proposal #2: UniMart

Campus-Based Digital Marketplace System

This project aims to develop a secure, campus-based digital marketplace where students can buy and sell products within their own university. The system supports two universities and uses official university email verification to ensure that only legitimate students can access the platform. To maintain security and simplicity, transactions are restricted within the same university.

Project Objectives

- To develop a digital marketplace exclusively for university students.
- To ensure secure user authentication using official university email addresses.
- To enable buying and selling of products within the same campus.

Project Objectives

- To reduce trust and security issues found in general online marketplaces.
- To design a scalable system that can support more universities in the future

Literature Review

1. General Online Marketplaces
   a. Facebook Marketplace
   b. Campus Buy & Sell Facebook Groups

2. E-commerce Platforms
   a. OLX
   b. Daraz
   c. Bikroy.com

User Roles

- **Student (Buyer):**
  Can browse products, place orders, and communicate with sellers within the same university.

- **Student (Seller):**
  Can list products, set prices, and manage sales within their university marketplace.

- **Admin:**
  Responsible for user verification, monitoring activities, and handling reports or disputes.

Core Features

1. User Registration and Authentication

- Students must register using official university email
- System verifies email domain to ensure valid university access
- Secure login and logout functionality

2. Role-Based User Access

- Single student account can act as Buyer or Seller
- Admin has separate access with higher privileges

3. Student Profile Management

- Basic profile information (Name, University, Student ID)
- Automatic university identification from email.
- Profile update functionality

4. Product Browsing (Buyer)

- Buyers can browse products within their own university
- Product search by name
- Filter products by category
- View detailed product information

5. Product Listing System (Seller)

- Sellers can add products with:
  - Product name
  - Description
  - Price
  - Category
  - Condition (New / Used)

- Sellers can edit or delete their own product listings Unauthorized users are restricted from accessing the system

6. Order Placement System

- Buyers can place orders for selected products
- Orders are restricted **within the same university**
- Order status starts as Pending

7. Order Management (Seller)

- Sellers can view incoming orders
- Sellers can Accept or Reject orders
- Order status updates accordingly

8. Service Fee Logic

- System automatically applies a fixed service fee per order
- Fee is shown clearly during order placement
- Ensures transparency in transactions

9. Reporting System

- Buyers can report suspicious or inappropriate product listings
- Reports include reason and description
- Helps maintain trust and safety

10. Admin Management Panel

- Admin can view users, products, and orders
- Admin can remove reported or inappropriate listings
- Admin monitors system activity and handles reports

Thank You